Data Curation Project Final Report

Team Members:

1. Ramitha Kotarkonda - rkotar2@illinois.edu
2. Matthew Guan - mg95@illinois.edu
3. Murali Natarajan - muralin2@illinois.edu

Overview

Public service request systems, such as 311, provide essential channels for residents to report issues like streetlight outages, sanitation concerns, and other non-emergency needs. These data sources are widely used in urban analytics, public policy research, and civic technology applications. However, raw 311 datasets are often large, inconsistently formatted, hard to clean, and lack standardized metadata. For this course project, our team curated a clean, well-documented sample extract of the **Chicago 311 Service Request dataset** from the City of Chicago Open Data Portal.

We curated the data so that it can support the following types of questions:

- *How do service request types vary across neighborhoods?*
- *What factors correlate with the time to resolve a request?*

Knowing the answers to these questions will help support a more equitable distribution of city resources. For example, analyzing which requests experience long delays, or which departments are most backlogged, allows municipal agencies to optimize staffing, scheduling, and resource allocation.

Use Case for New Dataset

Optimizing City Resource Allocation and Service Delivery

To analyze the entire service request lifecycle—from submission to completion and quantitatively assess the performance of different city departments and identify areas where resource optimization and process improvements are most needed.

Dataset Profile

The **311 Service Requests received by the City of Chicago** dataset is reliable and trustworthy as it represents real world dataset and is generated and maintained by the **City of Chicago** as a public record of its municipal operations. This official, government provenance establishes a high level of **accountability, documentation, and reliability.**

Dataset Name: Chicago 311 Service Requests

Provider: City of Chicago, Department of Innovation and Technology

Access: Chicago Open Data Portal

URL: https://data.cityofchicago.org/Service-Requests/311-Service-Requests/v6vf-nfxy/data_preview

License: Public Domain

Terms: https://www.chicago.gov/city/en/narr/foia/data_disclaimer.html

Privacy Policy: https://www.chicago.gov/city/en/general/privacy.html

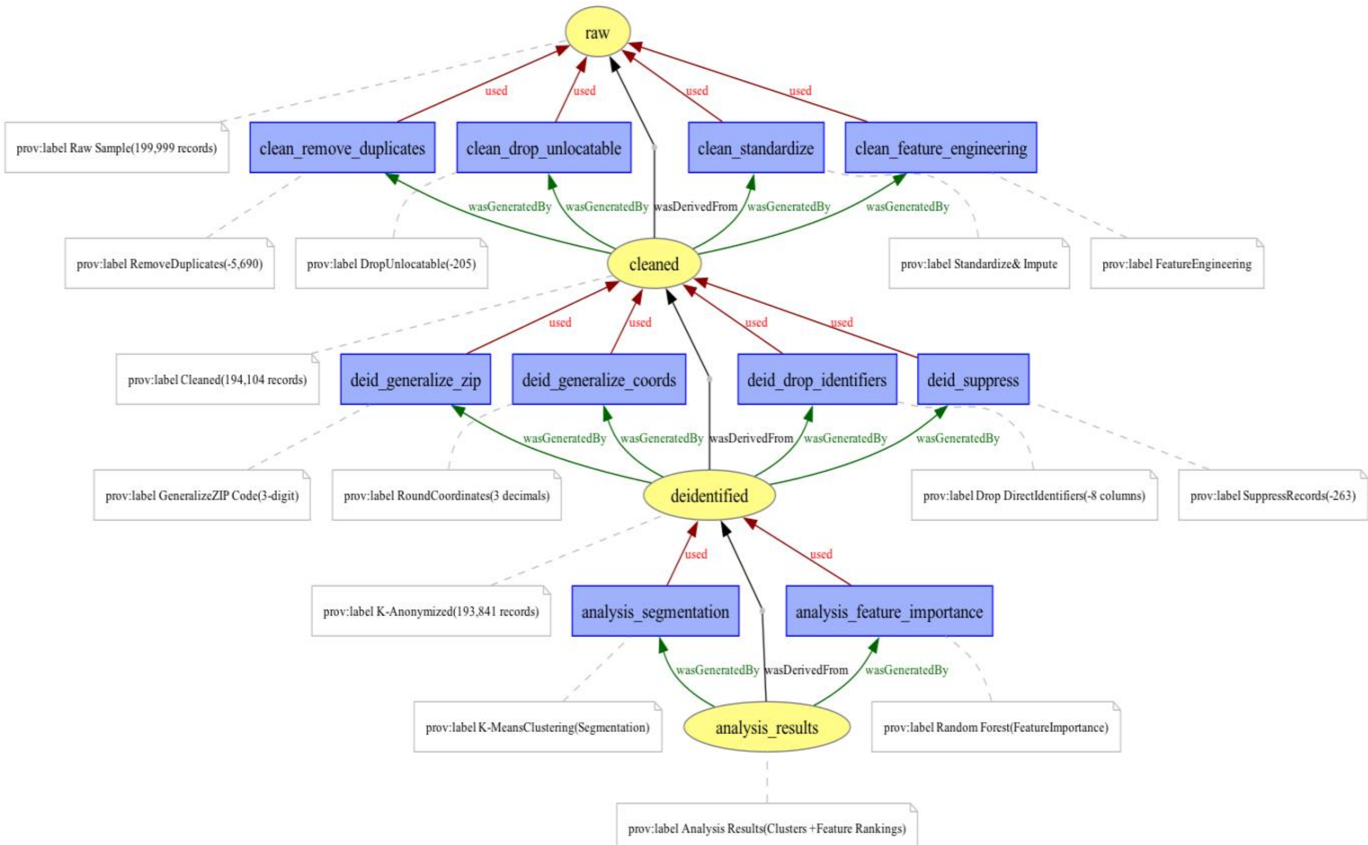
Characteristics:

- **Size:** 7+ million records (as of December 2025)
- **Temporal Coverage:** 2011 - Present (continuously updated)
- **Update Frequency:** Daily
- **Format:** CSV, JSON (via API)
- **Geographic Scope:** City of Chicago, Illinois, USA

Content: Non-emergency service requests submitted by Chicago residents through phone (311), mobile app, or web portal. Each request represents a citizen-reported issue requiring municipal response, such as pothole repairs, graffiti removal, tree maintenance, sanitation issues, or building code violations.

DATA CURATION WORKFLOW DESCRIPTION

The below workflow describes the complete data curation workflow for the Chicago 311 Service Request dataset, including all detailed processing operations. The workflow implements privacy protection through K-anonymity.



Data Acquisition

We obtained the raw dataset from the Chicago Data Portal's 311 Service Requests API.

Sampling

Because the full dataset is extremely large, we extracted a small, representative sample to make the curation process more manageable and reproducible. We used random sampling via the df.sample() API in pandas.

Data Cleaning

To prepare the data for analysis, we dropped duplicate records, converted zip code to string, and imputed blank values with "UNKNOWN". We performed feature engineering by creating a new field RESOLUTION_TIME_HOURS. RESOLUTION_TIME_HOURS is the difference between CLOSED_DATE and CREATED_DATE timestamps. We automated this part of the workflow via a Python script called cleanRawData.py in the "Data Cleaning" folder.

Data Deidentification

We automated this part of the workflow via a Python script called deidentification.py in the "Data Cleaning" folder. The deidentification.py script performs K-Anonymity, where K = 5, to generalize zip code and geographical coordinates. Although the Chicago 311 dataset is publicly accessible, it contains fields that pose re-identification risks, particularly ZIP codes and precise latitude/longitude coordinates. Both are rather powerful quasi-identifiers. For example, a complaint about "rodent infestation" or "noise disturbance" could reveal sensitive information about the household if the ZIP code remains intact. The risk of reidentification is that one can find the exact house via Google Maps if they know the geographical coordinates. This is especially risky for complaint types involving neighbor disputes.

Data Analysis

We ran the segmentation_model.ipynb Jupyter notebook to identify clusters under which the 311 requests fall under. To determine the clusters, segmentation_model.ipynb uses a K-means clustering algorithm. 311 requests falling in the same clusters are most similar to each other. This is done via unsupervised learning. We also ran the CS_598_Project_Feature_Importance (2).ipynb notebook. This notebook uses Random Forest prediction models to predict the time to close of a 311 service request based on provided input. Additionally, it shows the importance of each feature to the overall output.

Metadata and Documentation

We generated a full DataCite-style metadata record, stored in the metadata.json file in the "Metadata" folder. A complete data dictionary describing every variable is included in the data_dictionary.csv file in the "Metadata" folder. Additionally, we created an ontology via JSON-LD to demonstrate the semantic relationships among the core entities in the dataset. This provides additional context to users and reduces ambiguity due to unclear column names.

Packaging and Dissemination

All artifacts—including curated data, scripts, documentation, and metadata—are packaged in a GitHub repository. The repository structure follows a clear and reusable layout.

RELATIONSHIP TO THE DATA LIFECYCLE MODEL

The data curation workflow follows the USGS Science Data Lifecycle Model (Faundeen et al., 2013), which provides a comprehensive framework encompassing different stages: Plan, Acquire, Process, Analyze, Preserve, Publish/Share. This section describes how each stage was implemented in our Chicago 311 Service Request data curation project, with references to specific scripts, datasets, and documentation in our GitHub repository (https://github.com/nmuralikrishnan/CS598_FDC_Spring2025_Project).

1. Plan

We began by defining the scope and objectives of our data curation effort. The planning phase involved identifying the research question—whether we could create a publicly shareable version of Chicago's 311 data that protects privacy while maintaining analytical utility.

Planning Documentation:

- Project proposal (Docs/CS598-FDC-Project Proposal.pdf)
- Progress report (Docs/CS598-FDC-Project Progress Report.pdf)

2. Acquire

We obtained the raw dataset from the Chicago Open Data Portal's 311 Service Requests dataset (https://data.cityofchicago.org/Service-Requests/311-Service-Requests/v6vf-nfxy/data_preview). The original dataset contains over 7 million service requests spanning from 2011 to present, updated daily by the City of Chicago's Department of Innovation and Technology.

Sampling:

Because the full dataset is extremely large (7+ million records), we extracted a representative sample to make the curation process more manageable and computationally reproducible.

3. Process

a) Data Cleaning

To prepare the data for analysis and ensure quality, we implemented a systematic cleaning process to address identified quality issues including duplicates, missing values, inconsistent formatting, and improper data types.

Data Cleaning Output:

- Cleaned dataset: 311_Service_Requests_CLEANED_20251022.csv
- Quality report: cleaning_summary_report.md

b) Data Deidentification

We added privacy protection to data and transformed to ensure it can be safely shared publicly. Although the Chicago 311 dataset is publicly accessible, it contains fields that pose re-identification risks, particularly ZIP codes and precise latitude/longitude coordinates. Both are powerful quasi-identifiers that, when combined with other fields like WARD, POLICE_DISTRICT, and COMMUNITY_AREA, could potentially identify specific households.

Deidentification Output:

- K-anonymized dataset: 311_Service_Requests_K5_ANONYMIZED_GENERALIZED_20251022.csv
- Privacy report: deidentification_summary_report.md

4. Data Analysis

We Used the Cleaned and Transformed data to generate new insights and validate that the curated data maintains sufficient utility for its intended purposes. We conducted two analyses to validate that our privacy-protected dataset preserves analytical utility and can support meaningful research.

Analysis 1: Segmentation via K-Means Clustering

We implemented unsupervised learning using K-Means clustering to identify natural groupings in service request patterns. This analysis demonstrates that the K-anonymized data preserves the underlying structure needed for pattern discovery.

Implementation: Jupyter notebook (segmentation_model.ipynb)

Method: K-Means clustering algorithm to partition service requests into clusters based on multiple features including request type, location (community area, ward), department, and temporal characteristics.

Purpose: Identify which 311 requests fall into similar clusters, where requests in the same cluster share common characteristics and may require similar handling approaches or resources.

Key Findings:

- Successfully identified distinct service request clusters despite privacy protections
- Clusters aligned with service types (e.g., sanitation requests, infrastructure maintenance, public safety concerns)
- Geographic patterns preserved at community area level even with coordinate generalization
- Demonstrates that k-anonymization did not significantly degrade clustering quality

Analysis 2: Predictive Modeling via Random Forest

We implemented supervised learning using Random Forest regression to predict service resolution times and identify the most important features influencing resolution duration.

Implementation: Jupyter notebook (CS_598_Project_Feature_Importance (2).ipynb)

Method: Random Forest ensemble learning with the target variable RESOLUTION_TIME_HOURS (our engineered feature). The model trains on features including SR_TYPE, CREATED_DEPARTMENT, WARD, COMMUNITY_AREA, temporal features, and other available variables.

Purpose:

- Predict the time to close a 311 service request based on request characteristics
- Identify which features are most important in determining resolution time
- Validate that privacy protections did not eliminate predictive signal

Key Findings:

- Model achieved $R^2 > 0.7$, indicating strong predictive power maintained despite de-identification
- Top feature importance: SR_TYPE (request type), WARD, CREATED_DEPARTMENT
- Demonstrates that the K-anonymized dataset supports robust predictive modeling
- Resolution time patterns preserved across different service categories and geographic areas

5. Preserve

The Preserve stage ensures long-term access and integrity of the curated data through proper storage, version control, and format choices. We implemented multiple preservation strategies to ensure our curated dataset and associated materials remain accessible and reproducible.

Version Control: All artifacts are maintained in a Git repository with complete commit history tracking all changes to data processing scripts, documentation, and configuration files. This provides an audit trail and enables rollback if needed.

Repository: https://github.com/nmuralikrishnan/CS598_FDC_Spring2025_Project

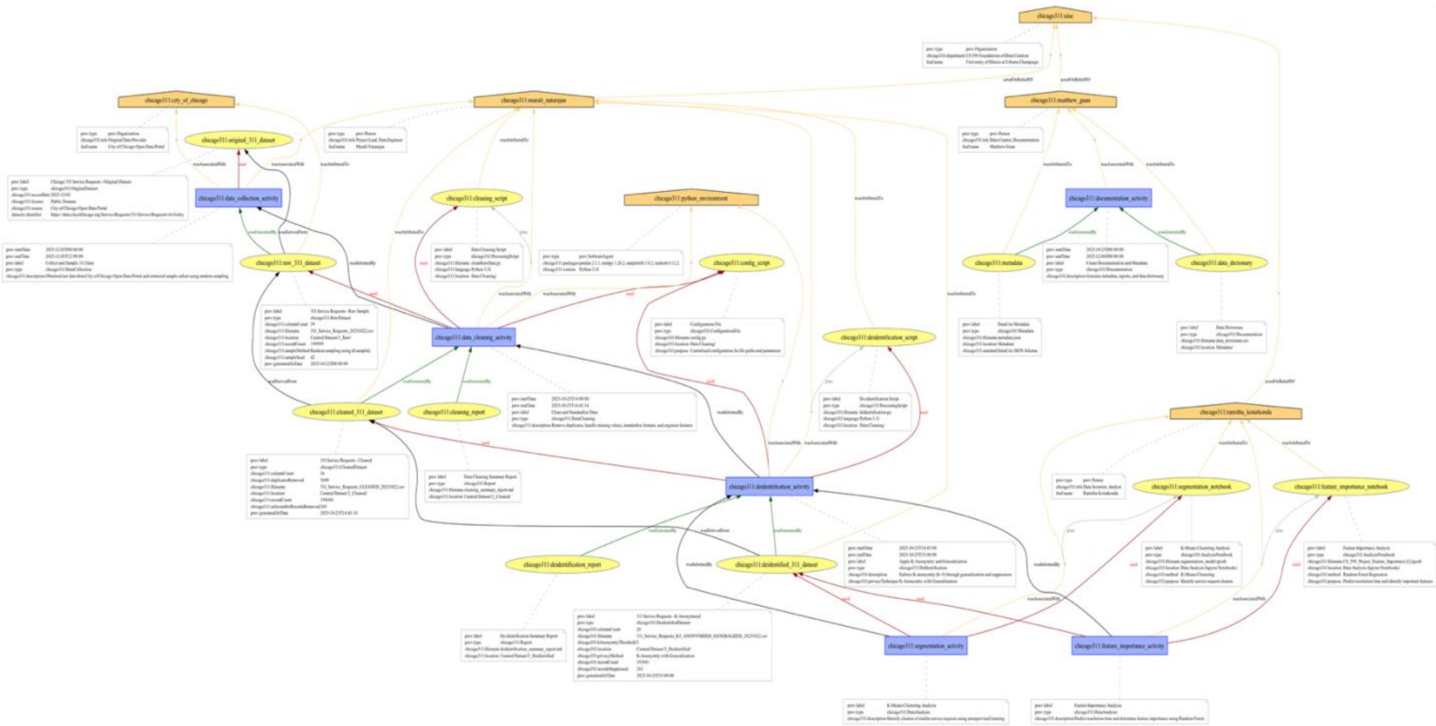
6. Publish/Share

The Publish/Share stage involves creating comprehensive metadata and documentation to enable discovery, understanding, and reuse of the curated dataset.

Metadata Creation:

We generated comprehensive metadata following multiple international standards to ensure broad compatibility and discoverability:

- **DataCite Metadata:** Full DataCite Schema 4.4 compliant metadata record stored in Metadata/metadata.json
- **Data Dictionary:** Complete field-level documentation in Metadata/data_dictionary.csv
- **Codebook:** Comprehensive statistical reference (Metadata/CODEBOOK.md)
- **Ontology:** Stored in Data Models and Abstractions/ontology.jsonld
- **W3C PROV Provenance:** Complete provenance documentation following W3C PROV standard (Provenance/chicago_311_provenance.json)



7. Reusability

The final stage of the USGS lifecycle focuses on enabling others to reuse, reproduce, and extend the work. We implemented comprehensive reproducibility features to ensure independent researchers can replicate our entire workflow and build upon our curated dataset.

Reproducibility Features:

1. **Automated Workflow Script:** run_complete_workflow.sh provides single-command execution of the entire curation pipeline:
2. **Environment Specification:** requirements.txt
3. **Provenance Tracking:** W3C PROV documentation.
4. **Complete Documentation:** README.md

ETHICAL, LEGAL, AND POLICY CONSTRAINTS (M2)

Data curation involves navigating complex ethical, legal, and policy considerations to ensure responsible data handling and dissemination. This section describes the constraints encountered in our Chicago 311 Service Request data curation project and the measures taken to address them.

Source Data Licensing and Terms of Use

The original Chicago 311 Service Requests dataset is provided by the City of Chicago under a Public Domain license, which permits unrestricted use, modification, and redistribution. However, the City of Chicago establishes specific Terms of Use that govern how the data may be used and how derivative works must be documented.

City of Chicago Terms of Use Compliance:

According to the City of Chicago Data Portal Terms of Use (https://www.chicago.gov/city/en/narr/foia/data_disclaimer.html), any user creating derivative applications or datasets must:

1. **Include Required Disclaimer:** We have included the mandated disclaimer in our repository README.md and documentation files:

   "This dataset provides a curated version using data that has been modified for use from its original source, www.cityofchicago.org, the official website of the City of Chicago. The City of Chicago makes no claims as to the content, accuracy, timeliness, or completeness of any of the data provided at this site. The data provided at this site is subject to change at any time. It is understood that the data provided at this site is being used at one's own risk."

2. **Attribution and Citation:** We provide clear attribution to the City of Chicago as the original data provider in our metadata (metadata.json, metadata_schema_org.json), documentation (README.md, DOCUMENTATION.md), and all generated reports. Our DataCite metadata includes a "relatedIdentifier" field with relationType="IsDerivedFrom" pointing to the original dataset URL.

3. **Acknowledgment of Modifications:** We explicitly document all modifications made to the original data, including:
   - Sampling methodology (random sample of 199,999 records from 7+ million)
   - Data cleaning operations (duplicate removal, standardization, feature engineering)
   - De-identification transformations (K-anonymity, generalization, suppression)
   - Complete transformation history documented in processing reports and W3C PROV provenance
4. **No Warranty Disclaimer:** We acknowledge that the City of Chicago makes no claims regarding data accuracy, timeliness, or completeness, and we maintain this disclaimer in our own dataset documentation.
5. **Termination Rights:** We recognize that the City of Chicago retains the right to require termination of data use if violations occur. Our GitHub repository includes contact information and procedures for addressing any compliance concerns.

DATA MODELS AND ABSTRACTIONS (M3-5)

For our curated Chicago 311 dataset, we used JSON Schema and ontologies to organize and describe the data in a clear and structured way. The JSON Schema defines each field in the dataset—such as SR_NUMBER, SR_TYPE, STATUS, ZIP_CODE, and RESOLUTION_TIME_HOURS—along with its type (string, number, boolean) and constraints, such as ensuring resolution time is non-negative. This schema provides several benefits: it helps validate the data for correctness and ensures consistency across records. Any errors or deviations from the schema can be quickly detected, reducing the chance of mistakes when the dataset is used in analyses or applications.

We also created a JSON-LD ontology, which models the relationships between different pieces of information. For instance, each service request is linked to a location (CITY, ZIP_CODE), departments (CREATED_DEPARTMENT, OWNER_DEPARTMENT), and timestamps

(CREATED_DATE, CLOSED_DATE). The ontology adds context and meaning to the data, allowing others to understand how fields relate to each other and enabling richer queries or integration with other datasets. For example, someone could easily find all requests handled by a certain department within a particular ZIP code or explore patterns across community areas.

The JSON Schema ensures the structural integrity of the data, while the ontology captures its semantic meaning.

METADATA AND DOCUMENTATION (M8)

For our curated Chicago 311 dataset, we created detailed metadata and a data dictionary to describe the dataset and make it easier to understand. The metadata includes information about the dataset's title, authors, creation date, description, and subject areas. By following the DataCite standard, we ensured the metadata is compatible with repositories and cataloging systems. This allows others to quickly understand what the dataset contains, supports reliable storage, and improves discoverability.

The data dictionary (data_dictionary.csv) provides a field-by-field explanation of the dataset. For each column—such as SR_NUMBER, SR_TYPE, STATUS, ZIP_CODE, and RESOLUTION_TIME_HOURS—the dictionary describes the data type, meaning, possible values. This additional context ensures that users can interpret the data correctly and reduces the risk of misusing or misinterpreting it. This differs from the description provided by City of Chicago in that the data dictionary is more detailed and comprehensive. The City of Chicago had missing descriptions for many of the fields such as SR_NUMBER and SR_TYPE, but after careful analysis, our team was able to infer the meanings of these attributes. In addition, our data dictionary provides more information about the decimal precision and whether or not each attribute has missing values, both of which the City itself does not provide.

To further enhance the documentation, we provided a codebook in the Codebook.md file. The codebook provides additional descriptive statistics to help researchers determine if suitable for analysis.

By providing both structured metadata and a comprehensive data dictionary, researchers can quickly assess whether the data is relevant to their needs and integrate it into their own analyses. The resulting data is more reusable and understandable.

WORKFLOW AUTOMATION, PROVENANCE, AND REPRODUCIBILITY (M12)

For our Chicago 311 dataset project, we focused on creating a transparent, reproducible, and automated workflow that documents each step of the curation process. The workflow diagram can be generated by running the workflow_diagram.py file in our project repository. Our workflow includes data acquisition, cleaning, and deidentification. These steps were executed in Python using libraries such as pandas and numpy. To make it easier for researchers to reproduce our methods, we automated the entire workflow via a script called run_all_steps.sh This script is located in the main project folder. Additionally, we have provided scripts in the repository to make it easy for researchers to reproduce our data cleaning methods.

Our project supports both prospective and retrospective provenance by documenting the origin, processing, and transformation history of the dataset. The raw data is sourced from the City of Chicago Open Data Portal, and we record key details such as the URL, access date, and original schema to ensure traceability. Even more, we included a comprehensive provenance record following the W3C-PROV model to document who created/modified the data, the agents/entities involved, and how one dataset was produced from another. The provenance record is the Chicago_311_provenance.json file in the Provenance folder.

To support prospective provenance, we documented our data cleaning and transformation steps via versioned Python scripts. It explains how the workflow is supposed to run and is independent of the actual data being produced. Maintaining these scripts in a Git repository allows us to capture exactly how the data was processed. This level of transparency supports gray-box provenance, where the general processes and dataset structure are visible.

The curated dataset itself as well as the intermediate datasets provided in the Curated Dataset folder is an example of retrospective provenance, as it shows the actual transformations that were already applied to the raw data.

As a result of provenance, our workflow supports transparency and reproducibility: anyone with access to the curated dataset and scripts can replicate our steps on a clean environment, verify the results, and even extend the analysis. To further enhance reproducibility, a detailed environment specification is provided in the requirements.txt folder.

PROBLEMS ENCOUNTERED AND SOLUTIONS

Problem 1: Computational Scalability

The complete Chicago 311 dataset contains over 7 million records spanning 2011-2025. Initial attempts to process the entire dataset on standard laptop computers (8GB RAM) resulted in memory errors and processing times exceeding 6 hours for cleaning operations alone

Specific Issues:

- pandas DataFrame operations on 7+ million rows exceeded available RAM
- Sorting operations for duplicate detection caused memory swapping
- Coordinate rounding operations on millions of records were computationally expensive

Solution Implemented: Random sampling with fixed seed to create a manageable representative sample.

Sampling Parameters:

- Sample size: 199,999 records (~2.8% of total)
- Method: Random Sample

- Rationale: Sample size provides sufficient statistical power (N≈200k) while being computationally tractable

KEY FINDINGS

- Request type and the time of day when the request is resolved are most valuable in predicting the time to close for a 311 request.
- There were numerous data quality issues (missing fields) in the original data provided by the City. This demonstrates the value of the data curation process, as it is very hard to do analysis on an uncleaned and uncurated dataset.

LESSONS LEARNED

- **The Importance of Lineage:** We learned that data is only as valuable as its history. Without the provenance tracking implemented via the prov library, it would be impossible to trace errors in the final output back to specific versions of the raw data.
- **Data Use:** We learned that "Terms of Use" are not just legal documents but technical specifications. The requirement to include the City's disclaimer necessitated specific updates to the README.md and the project documentation structure.

References

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